PS-41 Geotechnical Borehole and Groundwater Summary

Brookside Road pump-station excavation - source record 003

1. Borehole and Subsurface Summary

The borehole information is a project-specific summary for constructability planning. It is not a geotechnical design report and must be read with the temporary works design records supplied separately.

Borehole	Station / offset	Ground el. m	Key strata	Groundwater observation	Planning note
BH-01	5+057, 2.5 m LT	102.54	0.18 m asphalt; fill to 100.90; silty sand to 98.80; stiff clay below	Seepage at 100.72 after 2 h	West side may seep before brace level 1.
BH-02	5+073, centerline	102.61	0.16 m asphalt; fill to 101.05; loose sand/silt to 98.20; clayey silt to 96.20; shale below	Static water at 100.15 after 24 h	Water table is 3.75 m above design bottom 96.40.
BH-03	5+089, 2.0 m RT	102.48	0.20 m asphalt; fill to 100.70; sandy silt to 98.55; stiff clay below	Seepage at 100.42 after 3 h	East side inflow expected near utility crossings.

2. Groundwater and Excavation Inputs

Record ID	Parameter	Supplied value	Unit	Use in planning
GW-001	Design excavation bottom elevation	96.40	m	Reference for dewatering and sheet-pile toe checks
GW-002	Representative static groundwater elevation	100.15	m	Groundwater is 3.75 m above bottom
GW-003	Expected steady inflow to braced box	32	L/s	Base inflow component for pump capacity
GW-004	Rainfall and seepage allowance	15	L/s	Additional inflow component before reserve factor
GW-005	Reserve factor for operating capacity	1.25	ratio	Used with GW-003 plus GW-004
GW-006	Calculated required operating capacity	58.75	L/s	(32 + 15) x 1.25
GW-007	Lower excavation threshold	100.60	m	Dewatering must operate before excavation below this elevation
GW-008	Maximum drawdown at building wells	1.20	m	Monitoring trigger for adjacent building wells
GW-009	Maximum settlement trigger	8.0	mm	Settlement action threshold
GW-010	Vibration trigger	5.0	mm/s	Trigger during sheet-pile driving

3. Monitoring Network

Monitor ID	Location	Purpose	Baseline supplied	Trigger	Comments
MW-01	North curb, station 5+060	Groundwater drawdown	Yes	1.20 m drawdown	Baseline appears usable.
MW-02	Residential wall, station 5+075	Groundwater drawdown	No	1.20 m drawdown	Baseline missing; monitoring trend cannot be calculated.
MW-03	Shopfront wall, station 5+086	Groundwater drawdown	Yes	1.20 m drawdown	Closest building well to excavation.
SM-01	North sidewalk, station 5+071	Settlement point	Yes	8.0 mm settlement	Use during sheet pile driving and lower excavation.
VM-01	Gas valve chamber, station 5+072	Vibration	Yes	5.0 mm/s PPV	Use during sheet-pile driving near gas service.

4. Geotechnical Planning Constraints

Constraint ID	Constraint	Required project action	Planning risk
GT-001	Loose silty sand is present above clay in the excavation interval.	Maintain dewatering before and during excavation below elevation 100.60 m.	Loss of ground or running sand if dewatering is late.
GT-002	Observed water elevations are above excavation bottom.	Verify operating pump duty capacity before lower excavation.	Pump shortfall can stop excavation.
GT-003	Monitoring baseline at MW-02 is not supplied.	Obtain baseline before drawdown trend is used.	Drawdown cannot be verified at one residential wall.
GT-004	Settlement trigger is 8.0 mm.	Link settlement readings to daily excavation log.	Trigger exceedance requires project-engineer review.
GT-005	Vibration trigger is 5.0 mm/s.	Review sheet-pile driving method near gas and building monitors.	Trigger exceedance can delay sheet pile installation.